

- Q.28 Explain array implementation of queue.
- Q.29 What is meant by complete binary tree?
- Q.30 Discuss operations on binary search tree in brief.
- Q.31 Write a program to add two matrices using functions.
- Q.32 Write various operations which can be performed on primitive data structures.
- Q.33 How a two dimensional array can be implemented in memory?
- Q.34 Differentiate between linear list and doubly linked list?
- Q.35 Discuss various operations that can be performed on a queue.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Convert the following expression into post fix notation using stack
6 1 0 2 9 1 5 7
- Q.37 Explain the term sorting. What are various sorting methods used for internal sort? Discuss in brief.
- Q.38 Write short note on the following :
- doubly linked list
 - linear search

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4th Sem / Comp Subject:- Data Structures Using C

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A stack is a linear list in which insertion and deletion can be done through
- both ends
 - one end
 - both a and b
 - none of these
- Q.2 The operation of finding the location of an element in the array is called
- sorting
 - searching
 - merging
 - none of these
- Q.3 Two pointers used in queue operation
- front and rear
 - push and pop
 - postfix and prefix
 - left and right
- Q.4 Queue needs
- FIFO
 - LIFO
 - both a and b
 - none of these

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- Q.5 The process of accessing and processing of each element of the array exactly once is called as:
- inserting
 - searching
 - traversing
 - deleting
- Q.6 Info field of node contains
- root
 - Information
 - pointer to next node
 - none of these
- Q.7 The nodes where out degree is zero are called
- root
 - left subtree
 - right subtree
 - leaf
- Q.8 Prefix notation is also known as _____ notation.
- polish
 - Infix
 - postfix
 - statement
- Q.9 Dequeue is _____ Operation on queue.
- insertion
 - deletion
 - addition
 - modification
- Q.10 Smallest element in binary search tree is
- left most element
 - right most element
 - centre most element
 - none of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define data type.
- Q.12 One dimensional arrays are also known as _____.

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- Q.13 LIFO stands for _____.
- Q.14 When the function calls itself it is called _____.
- Q.15 Variable that stores the address of another variable is called _____.
- Q.16 Node of a linked list contains _____ & _____ parts.
- Q.17 The two types of search techniques are _____ & _____ searching.
- Q.18 Tree is a linear data structure. (True/False)
- Q.19 Node with no children is called leaf. (True/False)
- Q.20 A tree in which every node has an out degree exactly equal to zero or two is known as complete binary tree. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Give algorithm for deleting an element from the stack.
- Q.22 Differentiate between stack and queue.
- Q.23 Distinguish between linear queue and circular queue.
- Q.24 Define array. Give algorithm for traversing an array.
- Q.25 Write the applications of linked list.
- Q.26 Explain linear and non-linear data structure.
- Q.27 What is a dequeue? Explain its types in brief.

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